Functional Annotation Report: paaG (PP_3283, UniProt Q88HS0)

Pseudomonas putida KT2440 — ring-1,2-epoxyphenylacetyl-CoA isomerase (EC 5.3.3.18)

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1. Summary (Answer to the Research Question)

paaG encodes **ring-1,2-epoxyphenylacetyl-CoA isomerase** (also called oxepin-CoA–forming isomerase; EC 5.3.3.18), a **cytoplasmic**, crotonase-superfamily enzyme that catalyzes a key step of the **aerobic "hybrid" phenylacetate (PAA) degradation pathway**. Its primary reaction is the intramolecular isomerization of the reactive, non-aromatic **1,2-epoxyphenylacetyl-CoA** into the seven-membered oxygen-heterocycle **oxepin-CoA** (2-oxepin-2(3H)-ylideneacetyl-CoA). This step converts the toxic ring-epoxide produced by the PaaABC(D)E oxygenase into an α,β-unsaturated CoA-ester that is poised for hydrolytic ring cleavage (by PaaZ) and β-oxidation, ultimately yielding **acetyl-CoA and succinyl-CoA** that feed central metabolism. The protein is a soluble homotrimer of the enoyl-CoA hydratase/isomerase (crotonase) fold and works together with the pathway's other β-oxidation-like enzymes (notably PaaF, with which it forms a PaaFG complex).

Gene-identity verification: The UniProt target (Q88HS0; gene *paaG*; locus PP_3283; family = enoyl-CoA hydratase/isomerase; domains ClpP/crotonase-like IPR029045, Enoyl-CoA hydratase/isomerase IPR001753, PaaB1 IPR011968, PF00378 ECH_1; EC 5.3.3.18) is **fully consistent** with the literature on bacterial PaaG. The landmark pathway paper explicitly states the pathway operates in *Pseudomonas putida* as well as *E. coli*, so mechanistic knowledge transfers directly to PP_3283. No gene-symbol ambiguity was encountered.

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2. Biological Context: The Phenylacetate (paa) Catabolon

Phenylacetic acid (PAA) is a central hub intermediate in the aerobic catabolism of many aromatic compounds — phenylalanine, 2-phenylethylamine, 2-phenylethanol, styrene, lignin-related aromatics, and various environmental pollutants all converge on PAA (Teufel et al. 2010, P 20660314; Ismail et al. 2003, P 12846838). In *P. putida* this is organized as the **phenylacetyl-CoA "catabolon"**: peripheral routes that convert diverse substrates to PAA, feeding a common core that degrades phenylacetyl-CoA (Olivera et al. 1998, P 9600981; García et al. 2000, P 11010921). The true inducer of the core pathway is **phenylacetyl-CoA itself** (García et al. 2000, P 11010921).

Unlike classical aerobic aromatic catabolism (which uses ring-hydroxylating dioxygenases and oxygenolytic ring cleavage of free acids), the PAA pathway is a **"hybrid" pathway**: all intermediates are handled as **coenzyme-A thioesters**, and ring cleavage is **non-oxygenolytic** (Fuchs 2008, P 18378589; Teufel et al. 2010, P 20660314). The pathway occurs in ~16% of sequenced bacterial genomes (Teufel et al. 2010, P 20660314).

Pathway sequence and PaaG's position

1. **PaaK** — phenylacetate–CoA ligase → **phenylacetyl-CoA**

2. **PaaABC(D)E** — multicomponent (di-iron) ring-1,2-monooxygenase/epoxidase → **1,2-epoxyphenylacetyl-CoA** (an unusual, reactive, non-aromatic ring epoxide; this enzyme is also a bifunctional oxygenase/deoxygenase that can reverse epoxidation to limit toxic epoxide build-up — Teufel, Friedrich & Fuchs 2012,
   P 22398448)
3. **PaaG (this gene) — ring-1,2-epoxyphenylacetyl-CoA isomerase → oxepin-CoA** ☆
4. **PaaZ** — bifunctional (R)-enoyl-CoA hydratase + $NADP^+$ aldehyde dehydrogenase → hydrolytic ring-opening then oxidation to **3-oxo-5,6-dehydrosuberyl-CoA** (Teufel et al. 2011,
   P 21296885)
5. **PaaF / PaaH / PaaJ** — enoyl-CoA hydratase / 3-hydroxyacyl-CoA dehydrogenase / thiolase (β-oxidation-like steps)
6. **Products: acetyl-CoA** + **succinyl-CoA** → TCA cycle (Teufel et al. 2010,
   P 20660314)

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3. Primary Function: The Reaction Catalyzed

3.1 Core reaction and substrate specificity

PaaG catalyzes:

> **1,2-epoxyphenylacetyl-CoA → oxepin-CoA (2-oxepin-2(3H)-ylideneacetyl-CoA)**

This is an **intramolecular isomerization** — the epoxide oxygen is incorporated into a ring-expanded seven-membered oxepin enol-ether without net change in molecular formula, consistent with its EC class **5.3.3 (intramolecular oxidoreductases, transposing C=C bonds)** and specific number **EC 5.3.3.18**.

- Teufel et al. (2010,
  P 20660314):

> *"The reactive nonaromatic epoxide is isomerized to a seven-member O-heterocyclic enol ether, an oxepin."*

- Spieker, Saleem-Batcha & Teufel (2019, 31689071):
  "1,2-epoxyphenylacetyl-CoA is converted by PaaG into the heterocyclic oxepin-CoA."

Substrate specificity: PaaG acts specifically on the CoA-thioester ring-epoxide of the PAA pathway; the CoA moiety and the aromatic-derived ring are both accommodated in a large active-site cavity (Kichise et al. 2009, 19452559). By generating an **α,β-unsaturated CoA-ester**, PaaG "predisposes" the molecule for the subsequent hydrolytic ring-cleavage and β-oxidation (Spieker et al. 2019, 31689071).

3.2 A verified second (moonlighting) isomerase role

Spieker et al. (2019, 31689071) *"verify that PaaG adopts a second role in aromatic catabolism"* — i.e., beyond forming oxepin-CoA it catalyzes a further isomerization step in the pathway, consistent with the enzyme's Δ^3,Δ^2-enoyl-CoA-isomerase-like active site. This dual functionality is efficient use of a single crotonase scaffold.

3.3 Physiological requirement (genetic evidence)

Precise single-gene genetics confirm the step: - In *E. coli* K-12, *"mutants with a deletion of the paaF, paaG, paaH, paaJ or paaZ gene are unable to grow with phenylacetate as carbon source"* (Ismail et al. 2003, 12846838). - A *paaG* mutant fed ^{13}C-phenylacetate accumulated **ring-1,2-dihydroxy-1,2-dihydrophenylacetyl lactone**, and *paaG/paaZ* mutants also formed **ortho-hydroxyphenylacetate** (a known dead-end product) — showing that in the absence of PaaG the reactive epoxide collapses non-enzymatically, and pinpointing PaaG immediately downstream of ring epoxidation (Ismail et al. 2003, P 12846838).

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4. Structure and Mechanism (Evidence from Structural Biology)

- **Fold/family:** PaaG is a member of the **crotonase (enoyl-CoA hydratase/isomerase) superfamily** — matching the Q88HS0 annotations (ClpP/crotonase-like IPR029045; Enoyl-CoA hydratase/isomerase IPR001753; PF00378). Kichise et al. (2009, P 19452559): *"A phenylacetic acid degradation protein PaaG is a member of the crotonase superfamily, and is a candidate non-oxygenolytic ring-opening enzyme."*
- **Quaternary structure:** homotrimer — *"PaaG consists of three identical subunits related by local three-fold symmetry"* (T. thermophilus PaaG, 1.85 Å; Kichise et al. 2009, P 19452559). Crotonase enzymes typically assemble as trimers/dimers-of-trimers (hexamers).
- **Active site / catalytic residue:** *"Asp136 is the only conserved polar residue in the active site"*, positioned analogously to the catalytic residues of 4-chlorobenzoyl-CoA dehalogenase and peroxisomal Δ^3,Δ^2-enoyl-CoA isomerase, *"indicating that PaaG may undergo isomerization or a ring-opening reaction"* (Kichise et al. 2009, P 19452559). The active-site cavity is large enough to accommodate the ring substrate, and helix H2 adopts alternative conformations, implying substantial conformational motion during catalysis.
- **Complex formation:** PaaG forms a **stable PaaFG complex** with PaaF: *"another stable complex, PaaFG, an enoyl-CoA hydratase and enoyl-Coa isomerase, both belonging to the crotonase superfamily"*, with *"a four-layered structure composed of homotrimeric discs of PaaF and PaaG"* (Grishin et al. 2012,

).

P 22961985

This physical coupling of consecutive β-oxidation-like steps may allow substrate channeling.

The catalytic strategy (crotonase oxyanion hole + a single acidic residue promoting enolate/carbanion intermediates) is well suited to opening the strained epoxide and re-forming the C–O ring as the oxepin enol-ether.

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5. Subcellular Localization

PaaG functions in the **cytoplasm (cytosol)**: - It acts on **CoA-thioester** substrates generated intracellularly (phenylacetyl-CoA is produced by cytoplasmic PaaK and is the pathway inducer; García et al. 2000,

P 11010921

). - The protein is a **soluble crotonase-fold homotrimer** with no signal peptide or transmembrane region, and it forms a soluble multi-enzyme assembly with PaaF (Grishin et al. 2012,

P 22961985

; Kichise et al. 2009,

P 19452559

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There is no evidence for membrane association or secretion; the entire PAA core pathway is a soluble cytoplasmic module.

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6. Pathway Significance and Downstream Connections

- **Detoxification / channeling:** By rapidly converting the reactive ring-epoxide to oxepin-CoA, PaaG prevents accumulation of the toxic epoxide and of dead-end products (2-hydroxyphenylacetate, dihydrodiol lactone). The upstream oxygenase (PaaABC(D)E) has its own "escape" deoxygenase activity to limit epoxide toxicity when downstream steps stall (Teufel, Friedrich & Fuchs 2012,

P 22398448),
underscoring how tightly the epoxide→oxepin transition (PaaG) is controlled.

- **Metabolic branch point / secondary metabolism:** Oxepin-CoA (the PaaG product) and the downstream reactive aldehyde are precursors of tropone/tropolone secondary metabolites, including **tropodithietic acid** (an antibiotic/quorum-sensing signal) and ω-cycloheptyl fatty acids (Teufel et al. 2011,
P 21296885 ;
Spieker et al. 2019,
P 31689071).
Thus PaaG activity connects primary aromatic catabolism to natural-product biosynthesis.
- **Environmental/biotech relevance in *P. putida*:** The PAA catabolon underlies degradation of styrene and phenyl-derived pollutants and is linked to biosynthesis of medium-chain-length polyhydroxyalkanoate (PHA) bioplastics in *Pseudomonas* (O'Leary et al. 2005,
P 16085828 ;
García et al. 1999,
P 10506180).

7. Supported vs. Refuted Hypotheses

Hypothesis	Verdict	Key evidence
paaG (Q88HS0/PP_3283) encodes ring-1,2-epoxyphenylacetyl-CoA isomerase (EC 5.3.3.18) producing oxepin-CoA	**Supported**	Teufel 2010 (P 20660314); Spieker 2019 (P 31689071)
PaaG belongs to the crotonase (enoyl-CoA hydratase/isomerase) superfamily; homotrimeric; single catalytic Asp	**Supported**	Kichise 2009 (P 19452559); UniProt domain annotation
PaaG is essential for growth on phenylacetate	**Supported**	Ismail 2003 (P 12846838)
PaaG acts in the cytoplasm on CoA-thioesters, in a PaaFG complex	**Supported**	Grishin 2012 (P 22961985); García 2000 (P 11010921)
PaaG has an additional (second) isomerase role in the pathway	**Supported**	Spieker 2019 (P 31689071)
PaaG is a membrane-bound or oxygenolytic ring-cleavage enzyme	**Refuted**	Ring cleavage is non-oxygenolytic and performed by PaaZ; PaaG is a soluble isomerase (Teufel 2010, P 20660314; Teufel 2011, P 21296885)
The gene symbol "paaG" is ambiguous for this protein	**Refuted**	Symbol, organism, family, domains and EC all coherent with the PAA-pathway PaaG literature

8. Limitations and Future Directions

- **Direct biochemistry is on orthologs, not PP_3283 itself.** The enzymatic and structural characterizations were performed in *E. coli*, *Thermus thermophilus*, *Azoarcus evansii*, and related systems; *P. putida* KT2440 PaaG function is inferred by strong orthology and the explicit inclusion of *P. putida* in the pathway paradigm (Teufel et al. 2010, P 20660314). Strain-specific enzymology (kinetics, kcat/KM, substrate scope) of Q88HS0 has not been individually reported.
- **No experimental 3D structure of Q88HS0.** Structural inferences rely on the *T. thermophilus* PaaG structure and the *E. coli* PaaFG complex; an AlphaFold model of Q88HS0 could confirm the crotonase fold, trimer interface, and Asp active-site geometry.
- **The precise nature of PaaG's "second role"** (the exact downstream isomerization substrate/product) merits closer reading of Spieker et al. 2019 full text.
- **Regulation specifics** (σ^{54}/PaaX control of the KT2440 paa operon) were not deeply examined here but are relevant to when PaaG is expressed (cf. O'Leary et al. 2011, P 21995721 for *P. putida* CA-3).

9. Key References

1. Teufel R, Mascaraque V, Ismail W, et al. **Bacterial phenylalanine and phenylacetate catabolic pathway revealed.** *PNAS* 2010. P 20660314. *(Landmark pathway elucidation; oxepin isomerization.)*
2. Spieker M, Saleem-Batcha R, Teufel R. **Structural and Mechanistic Basis of an Oxepin-CoA Forming Isomerase in Bacterial Primary and Secondary Metabolism.** 2019. P 31689071. *(Direct PaaG mechanism; second role.)*

3. Ismail W, El-Said Mohamed M, et al. **Functional genomics by NMR spectroscopy. Phenylacetate catabolism in Escherichia coli.** 2003.

   P 12846838

   (paaG deletion genetics; trapped intermediates.)

4. Kichise T, Hisano T, Takeda K, Miki K. **Crystal structure of phenylacetic acid degradation protein PaaG from Thermus thermophilus HB8.** 2009.

   P 19452559

   (Crotonase fold; homotrimer; Asp136.)

5. Grishin AM, et al. **Protein–protein interactions in the β-oxidation part of the phenylacetate utilization pathway: crystal structure of the PaaF–PaaG hydratase–isomerase complex.** 2012.

   P 22961985

6. Teufel R, Gantert C, Voss M, et al. **Studies on the mechanism of ring hydrolysis in phenylacetate degradation: a metabolic branching point.** 2011.

   P 21296885

   (PaaZ ring cleavage; oxepin fate; tropodithietic acid link.)

7. Teufel R, Friedrich T, Fuchs G. **An oxygenase that forms and deoxygenates toxic epoxide.** 2012.

   P 22398448

   (PaaABCE epoxidase/deoxygenase; epoxide toxicity.)

8. Olivera ER, et al. **Molecular characterization of the phenylacetic acid catabolic pathway in Pseudomonas putida U: the phenylacetyl-CoA catabolon.** 1998.

   P 9600981

9. García B, et al. **Phenylacetyl-CoA is the true inducer of the phenylacetic acid catabolism pathway in Pseudomonas putida U.** 2000.

   P 11010921

10. Fuchs G. **Anaerobic metabolism of aromatic compounds.** 2008.

   P 18378589

   (Hybrid pathway concept.)